# Chromatic Numbers of the Plane with Two Forbidden Distances: a Seven-Color Lower Bound and Smaller Six-Chromatic Graphs

Henry Bass  Claude (Anthropic Fable 5)*

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**Abstract**

Let $\chi\big(\mathbb{R}^2; D\big)$ denote the minimum number of colors required to color the plane so that no two points whose distance lies in $D$ receive the same color, and set $d = (\sqrt{6}+\sqrt{2})/2 = 2\cos(\pi/12)$, so that $d^2 = 2+\sqrt{3}$ and $d^2 - 1 = \sqrt{2}\,d = 1+\sqrt{3}$. Our main result is

$$\chi\big(\mathbb{R}^2; \{1, d\}\big) \;\geq\; 7$$

for arbitrary colorings, with no measurability assumed (the averaging uses an invariant mean on the amenable group $E(2)$, a theorem of ZFC): for every proper 6-coloring, invariant monochromatic-pair densities $p_{\sqrt{2}}$ and $p_{1+\sqrt{3}}$ would have to satisfy $p_{\sqrt{2}} + p_{1+\sqrt{3}} \geq 1$ (an 8-point pigeonhole gadget of Parts) and $p_{\sqrt{2}}, p_{1+\sqrt{3}} \leq 5/11$ (two explicit 12-point chain configurations whose long-range pairs land exactly on the forbidden distances via the identity $d^2 - 1 = \sqrt{2}\,d$), a contradiction since $1 \leq 10/11$ is false. The argument realizes an unpublished 2019 sketch of J. Parts; the contribution here is a fully explicit, SAT-certified finite form with substantially stronger constants ($5/11$ in place of $\frac{1}{2} - \frac{1}{188}$), each component carrying a DRAT proof trace checkable by an independent ∼1300-line verifier, and the full 32-point geometry certified in exact symbolic arithmetic over explicit quadratic towers. When this work was done (August 2026) the best published bound for two distances was 6; see the note added below for later developments.

We further prove, with exact-arithmetic and doubly SAT-certified verification: a 26-vertex 6-chromatic $\{1, d\}$-graph (the smallest known 6-chromatic two-distance graph for *any* ratio, improving 31 [Parts 2020] and, for this ratio, 117 [De Neve et al. 2025]); a 30-vertex 6-chromatic $\{1, \sqrt{3}\}$-graph (previously 33); the six-coloring continuum $X_6 \supseteq [0.354, 0.667504532081]$ (previously $[0.354, 0.657]$); the edge bound $e_5 \leq 2278$ for 5-chromatic unit-distance graphs via an edge-critical trimming of the 509-vertex record graph (previously 2406); and forced-structure constants for invariant 5-colorings of the plane, including $p_{\sqrt{3}} \geq 1/12$ by an automated linear-programming "density ledger" and the single-certificate bounds $p_{\sqrt{3}} \geq 1/28$, $p_d \geq 1/37$. Finally, we exhibit the first color-count-free finite density caps at radii relevant to five colors ($p_{1/\sqrt{3}} \leq 1/3$ by four points; $p_{(1+\sqrt{5})/2} \leq 2/5$ by the unit pentagon; a dense family of capped radii from odd star polygons), prove that pairwise-conflict mechanisms can never beat $1/2$ at the ledger anchors $\sqrt{3}$ and 2 (a Niven-type bipartiteness theorem), and derive a crisp win condition: any 6-chromatic $\{1, \sqrt{3}\}$-graph forcing unit-edge monochromatic density above $1/3$ under $\sqrt{3}$-proper 5-colorings would prove $\chi(\mathbb{R}^2) \geq 6$. All certificates and ∼60-line verifiers are included in the reproducibility bundle (Appendix A).

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# 1 Introduction

The Hadwiger–Nelson problem asks for the chromatic number of the plane, $\chi(\mathbb{R}^2) = \chi\big(\mathbb{R}^2;\{1\}\big)$: the least number of colors so that no two points at distance 1 share a color. After more than half a century at $4 \le \chi(\mathbb{R}^2) \le 7$, de Grey [2] proved $\chi(\mathbb{R}^2) \ge 5$ with a 1581-vertex unit-distance graph, launching the Polymath16 project [14] and a rapid sequence of smaller witnesses [8, 5, 13].

In the *two-distance* variant one forbids two distances simultaneously: $\chi\big(\mathbb{R}^2;\{1,d\}\big)$ is the chromatic number of the graph on $\mathbb{R}^2$ with edges between points at distance 1 *or* $d$. Finite 6-chromatic witnesses were known for exactly four ratios: the golden ratio $(1+\sqrt{5})/2$ (Huddleston 2008 [9], non-explicit; explicit graphs and the 31-vertex record by Parts [12]), $d = 2$ (Exoo–Ismailescu [6], reduced to 397 vertices by Parts), and $d = \sqrt{3}$ and $d = (\sqrt{6}+\sqrt{2})/2$ (De Neve, Vanden Kerchove, Colle, Tavernier, and Pickavet [4], with 33 and 117 vertices respectively). For no finite set $D$ of distances was a bound $\chi\big(\mathbb{R}^2;D\big) \ge 7$ published, although J. Parts sketched a probabilistic argument for exactly this in a 2019 Polymath16 comment [11].

*Note added (October 2026).* After this work was done, OpenAI [18] proved $\chi(\mathbb{R}^2) \ge 6$, so every two-distance chromatic number is now at least 6 and the win condition of Section 7 is moot; the explicit small 6-chromatic graphs of Section 5 remain records. The transfer of [18] extends to finite sets of algebraic distances (see Crocker [19] and our companion preprint [20]), giving $\chi\big(\mathbb{R}^2;D\big) \ge 1/m_D(\mathbb{R}^2)$; combined with Bukh's theorem this yields $\chi\big(\mathbb{R}^2;\{1,N\}\big) \ge 17$ for all sufficiently large $N$, with no explicit threshold. Theorem 1.1 remains the only bound $\chi\big(\mathbb{R}^2;D\big) \ge 7$ for an explicit two-distance set that is proved by a finite certificate.

This paper reports the results of an intensive computer-assisted session (2026-08-08/09). Throughout, $d := (\sqrt{6}+\sqrt{2})/2 = 2\cos(\pi/12) \approx 1.93185$, whose relevant algebra is

$$d^2 = 2+\sqrt{3}, \qquad d^2 - 1 \;=\; 1+\sqrt{3} \;=\; \sqrt{2}\,d. \tag{1}$$

**Theorem 1.1** (Main theorem; invariant-coloring framework)**.** $\chi\big(\mathbb{R}^2;\{1,d\}\big) \ge 7$. *More precisely, there is no proper* 6*-coloring* $c\colon \mathbb{R}^2 \to \{1,\dots,6\}$ *avoiding monochromatic pairs at distances* 1 *and* $d$*: for any such* $c$*, invariant monochromatic-pair densities* $p_{\sqrt{2}}(c)$ *and* $p_{1+\sqrt{3}}(c)$ *exist (via an invariant mean on the isometry group) and satisfy the incompatible system*

$$p_{\sqrt{2}} + p_{1+\sqrt{3}} \;\ge\; 1, \qquad p_{\sqrt{2}} \;\le\; \tfrac{5}{11}, \qquad p_{1+\sqrt{3}} \;\le\; \tfrac{5}{11}.$$

The three displayed inequalities come from three finite configurations ($8+12+12 = 32$ points in total), each verified by two independent SAT solvers and certified by DRAT proof traces (Section 3; Appendix A). The framework, its provenance in ZFC, and its honest limitations are discussed in Sections 2–3; a counting blueprint that would replace the framework by a single finite graph is given in Section 4.

The remaining results are classical finite statements with exact-arithmetic certificates:

**Theorem 1.2.**

*(a) There are* 26 *points in the plane spanning* 85 *pairs at distance exactly* 1 *or exactly* $d$ *whose* $\{1,d\}$*-graph has chromatic number* 6 *and is vertex-critical. This is the smallest known* 6*-chromatic two-distance graph for any distance pair (previously* 31 *[12]; for this pair,* 117 *[4]).*

*(b) There are* 30 *points spanning* 100 *pairs at distance exactly* 1 *or exactly* $\sqrt{3}$ *whose graph has chromatic number* 6 *and is vertex-critical (previously* 33 *[4]).*

(c) *The record* 509*-vertex* 5*-chromatic unit-distance graph [13, 3] contains an edge-critical spanning subgraph with* 2278 *edges that is not* 4*-colorable. Hence* $e_5 \le 2278$ *(previously* $e_5 \le 2406$ *[3]).*

(d) *In every isometry-invariant proper* 5*-coloring of the plane (unit distance forbidden), the monochromatic-pair densities satisfy* $p_{\sqrt{3}} \ge 1/28$ *and* $p_d \ge 1/37$ *(single-graph certificates via (a),(b)), and* $p_{\sqrt{3}} \ge 1/12$ *(automated density-ledger linear program). The best previously implicit constant, via Parts'* 397*-vertex* $\{1,2\}$*-graph, was of order* $1/550$*.*

| Quantity | Previous | Here | Status |
|---|---|---|---|
| Smallest 6-chromatic two-distance graph, any ratio | 31 [12] | **26** | DRAT-verified |
| … for $d = (\sqrt{6}+\sqrt{2})/2$ | 117 [4] | **26** | DRAT-verified |
| … for $d = \sqrt{3}$ | 33 [4] | **30** | DRAT-verified |
| $\chi\big(\mathbb{R}^2;\{1,(\sqrt{6}+\sqrt{2})/2\}\big)$ | $\ge 6$ [4]; $\ge 7$ sketched [11] | $\ge$ **7** | 3 DRAT certi (invariant frame |
| Edges of the 509-vertex 5-chromatic UD graph | $\le 2406$ [3] | $\le$ **2278** | DRAT-verified, critical |
| Forced density $p_{\sqrt{3}}$ in invariant 5-colorings | — | $\ge$ **1/12** | LP over Ma exact ledger |
| Forced density $p_d$ in invariant 5-colorings | $\sim 1/550$ implicit | $\ge$ **1/37** | MaxSAT-exact |
| Finite strict-half cap $p_r \le \frac{1}{2} - \varepsilon$, any $k \ge 5$ | none | $p_r \le$ **5/11** $(k = 6)$ | MaxSAT-exact DRAT |
| Color-count-free finite caps | none | $p_{1/\sqrt{3}} \le$ **1/3**, $p_{(1+\sqrt{5})/2} \le$ **2/5** | elementary proo |

**Attribution.** The $\chi \ge 7$ argument is, in its skeleton, due to Jaan Parts, who sketched it in 2019 in comment 24686 on the fifteenth Polymath16 thread [11] and, to our knowledge, never published it. Our contribution is (i) fully explicit finite certificates — the specific 8- and 12-point configurations and their SAT/DRAT verification — (ii) much stronger constants (5/11 versus $\frac{1}{2} - \frac{1}{188}$ from the plain zigzag counting we reconstructed from the sketch), and (iii) a choice-free finitization program (Section 4). The record graphs of Theorem 1.2(a,b) were seeded from the published coordinates of De Neve et al. [4], without which the search would not have started.

**Organization.** Section 2 sets up two-distance graphs and the invariant-coloring framework. Section 3 proves Theorem 1.1. Section 4 describes the counting blueprint toward a fully finite witness and the obstruction theorems that constrain it. Section 5 presents the record graphs with complete coordinate tables and verification recipes. Section 6 contains the unit-distance results (Theorem 1.2(c,d)) and the measured structure of 5-colorings. Section 7 develops finite caps valid for any number of colors, an impossibility theorem at the anchors $\sqrt{3}$ and 2, and the win condition for $\chi(\mathbb{R}^2) \ge 6$. Section 9 collects open problems. Appendix A documents the artifact bundle.

## 2 Preliminaries

### 2.1 Two-distance graphs

For a finite $P \subset \mathbb{R}^2$ and a set $D$ of positive reals, let $G(P;D)$ be the graph on $P$ with $uv$ an edge iff $|u - v| \in D$. We write $\chi\big(\mathbb{R}^2;D\big)$ for the chromatic number of the (infinite) graph $G(\mathbb{R}^2;D)$

and abbreviate $G(P) = G(P; \{1, d\})$ when $D = \{1, d\}$ is understood. By the de Bruijn–Erdős theorem (with choice), $\chi\big(\mathbb{R}^2; D\big) \geq k$ holds iff some finite $P$ has $\chi(G(P; D)) \geq k$; part of the interest of Theorem 1.1 is precisely that it does *not* yet come with such a finite witness (see Section 3.5).

All configurations in Sections 5 have coordinates in a real quadratic-tower field ($\mathbb{Q}(\sqrt{3}, \sqrt{5})$ or $\mathbb{Q}(\sqrt{3}, \sqrt{7})$) with denominators dividing 8, so squared distances are computed in exact integer arithmetic and edge membership is decidable with no floating point. The $\chi \geq 7$ configurations of Section 3 are likewise exact algebraic configurations, certified end-to-end by symbolic computation in explicit quadratic towers (Section 3.3).

### 2.2 The invariant-coloring framework

We use the probabilistic formulation of Hadwiger–Nelson-type problems developed by the Polymath16 project [14], Ágoston [1], and Gwyn–Stavrianos [7]. The isometry group $E(2) = \mathrm{Isom}(\mathbb{R}^2)$ is solvable, hence amenable, so it admits a left-invariant mean $\Lambda$ on $\ell^\infty(E(2))$ (this uses Hahn–Banach; it is a theorem of ZFC).

**Definition 2.1.** Fix any proper coloring $c$ of $G(\mathbb{R}^2; D)$ with $k$ colors (no measurability assumed) and fix an invariant mean $\Lambda$. For $r > 0$ define the *monochromatic-pair density*

$$p_r \;=\; p_r(c, \Lambda) \;=\; \Lambda_{g \in E(2)}\Big[\,\mathbf{1}\big(c(g \cdot (0,0)) = c(g \cdot (r,0))\big)\Big] \;\in [0,1].$$

By left-invariance, for *every* ordered pair $(x, y)$ with $|x - y| = r$ we have $\Lambda_g[\mathbf{1}(c(gx) = c(gy))] = p_r$. This yields the transfer principle we use throughout:

**Lemma 2.2** (Finite transfer)**.** *Let $Q \subset \mathbb{R}^2$ be finite with designated pairs $(a_1, b_1), \ldots, (a_m, b_m)$, $|a_i - b_i| = r_i$, and let $t_{\min}$ (resp. $t_{\max}$) be the minimum (resp. maximum), over all proper $k$-colorings of the finite graph $G(Q; D)$, of the number of monochromatic designated pairs. Then for every proper $k$-coloring $c$ of $G(\mathbb{R}^2; D)$ and every invariant mean $\Lambda$,*

$$t_{\min} \;\leq\; \sum_{i=1}^{m} p_{r_i}(c, \Lambda) \;\leq\; t_{\max}.$$

*Proof.* For each $g \in E(2)$, the restriction of $c$ to $gQ$ is a proper $k$-coloring of $G(gQ; D) \cong G(Q; D)$, so the number of monochromatic designated pairs of $gQ$ lies in $[t_{\min}, t_{\max}]$ pointwise in $g$. Apply $\Lambda$ and use linearity together with the paragraph above, which gives $\Lambda_g[\mathbf{1}(c(ga_i) = c(gb_i))] = p_{r_i}$. □

Both $t_{\min}$ and $t_{\max}$ are computed exactly by maximum satisfiability (MaxSAT) over the finite graph and certified as follows: the optimum is found by an exact MaxSAT solver (RC2 over CaDiCaL), and the corresponding decision statement ("a proper $k$-coloring with $\geq t_{\max}+1$, resp. $\leq t_{\min}-1$, monochromatic designated pairs exists") is refuted by CNF unsatisfiability with a DRAT proof trace checked by `drat-trim` [17]. Realizability of the optimum is witnessed by an explicit coloring.

### 2.3 Prior work used directly

De Grey's graph [2] and its successors [8, 5, 13] give $\chi(\mathbb{R}^2) \geq 5$; the 509-vertex record and its 2406-edge trim are from [13, 3]. The two-distance 6-chromatic graphs we improve are from

[12, 6, 4]. The probabilistic framework is from [14, 1], and the finite-graph transfer of badness bounds from [7]. The $\chi \geq 7$ sketch is Parts' [11]. Set-theoretic sensitivity of chromatic numbers of distance graphs is due to Shelah–Soifer [15]; general background is in Soifer [16].

## 3 Seven colors for two distances

Throughout this section $k = 6$, $D = \{1, d\}$, and $c$ is an arbitrary proper 6-coloring of $G(\mathbb{R}^2; \{1, d\})$, assumed to exist for contradiction; $p_r = p_r(c, \Lambda)$ for a fixed invariant mean. The two auxiliary distances are $\sqrt{2}$ and $1 + \sqrt{3}$ — not themselves forbidden, but tied to $\{1, d\}$ by the identity (1).

### 3.1 The V8 pigeonhole

Parts' *V8 gadget* consists of two concentric squares: the unit square $I = \{(\pm\frac{1}{2}, \pm\frac{1}{2})\}$ and the outer square $O = \{(\pm s, 0), (0, \pm s)\}$ with $s = d/\sqrt{2} = (1 + \sqrt{3})/2$.

**Lemma 3.1.** *$G(I \cup O; \{1, d\}) = K_8$ minus the perfect matching consisting of the four "diagonals": the two inner diagonals at distance $\sqrt{2}$ and the two outer diagonals at distance $2s = 1 + \sqrt{3}$.*

*Proof.* Direct computation. Inner sides have length 1; inner diagonals $\sqrt{2}$. Outer sides have squared length $2s^2 = 2 + \sqrt{3} = d^2$; outer diagonals $2s = 1 + \sqrt{3}$. For an inner corner and an outer vertex, e.g. $(\frac{1}{2}, \frac{1}{2})$ and $(s, 0)$: the difference is $(\frac{\sqrt{3}}{2}, \frac{1}{2})$ of length 1; and against $(-s, 0)$ the squared distance is $\left(\frac{2+\sqrt{3}}{2}\right)^2 + \frac{1}{4} = 2 + \sqrt{3} = d^2$. By symmetry each inner vertex is at distance 1 from the two nearest outer vertices and $d$ from the two farthest. Thus all 24 pairs except the four diagonals are edges. □

**Proposition 3.2** (V8 pigeonhole)**.** *Every proper 6-coloring of $G(I \cup O; \{1, d\})$ makes at least 2 of the four diagonal pairs monochromatic. The bound is exact, and all 11 diagonal patterns of weight $\geq 2$ are realizable. Certificate: `v8cert.cnf` (proper 6-coloring $\wedge$ at most one monochromatic diagonal; 84 variables, 267 clauses) is UNSAT, with DRAT trace `v8cert.drat` verified by `drat-trim`.*

Intuitively: six colors on $K_8$ minus a perfect matching leave at most six "private" colors, so at least two matched pairs must share. (The pigeonhole $8 - 6 = 2$ is tight because the four diagonal pairs are disjoint.) By Lemma 2.2 with the four diagonals as designated pairs (two at $\sqrt{2}$, two at $1 + \sqrt{3}$) and $t_{\min} = 2$:

**Corollary 3.3.** $2p_{\sqrt{2}} + 2p_{1+\sqrt{3}} \geq 2$, *i.e.* $p_{\sqrt{2}} + p_{1+\sqrt{3}} \geq 1$.

### 3.2 The chain configurations and the exact identity

The counterweight to Corollary 3.3 is a pair of 12-point configurations bounding $p_{\sqrt{2}}$ and $p_{1+\sqrt{3}}$ from above. Fix a radius $r \in \{\sqrt{2}, 1 + \sqrt{3}\}$ and let

$$\varphi_1 = 2 \arcsin \frac{1}{2r}, \qquad \varphi_d = 2 \arcsin \frac{d}{2r}$$

be the angles subtending chords of length 1 and $d$ on a circle of radius $r$; equivalently $\cos\varphi_1 = 1 - \frac{1}{2r^2}$ and $\cos\varphi_d = 1 - \frac{d^2}{2r^2}$. Concretely,

$$r = \sqrt{2}: \quad \cos\varphi_1 = \tfrac{3}{4}, \quad \cos\varphi_d = \tfrac{2-\sqrt{3}}{4}; \qquad r = 1 + \sqrt{3}: \quad \cos\varphi_1 = \tfrac{2+\sqrt{3}}{4}, \quad \cos\varphi_d = \tfrac{3}{4},$$

a pleasing duality: the two radii exchange the roles of the two cosines.

Set $A = \varphi_d - \varphi_1$ and $B = \varphi_d + \varphi_1$, and define the *mixed chain* $X_0, \dots, X_6$ by $X_0 = (0,0)$ and $X_{i+1} = X_i + r(\cos h_i, \sin h_i)$, where the heading sequence has period four:

$$(h_0, h_1, h_2, h_3) \;=\; \big(0,\ \pi - A,\ -(A+B),\ \pi - B\big), \qquad h_{i+4} = h_i.$$

Thus consecutive chain points are at distance exactly $r$. The key fact is that the four-step displacement lands *exactly* on a forbidden distance:

**Lemma 3.4** (Four-step closure)**.** *For all $i$, $X_{i+4} - X_i$ is the same vector, of length $(d^2-1)/r$. Consequently, by (1): for $r = \sqrt{2}$, $|X_{i+4} - X_i| = (1+\sqrt{3})/\sqrt{2} = d$; for $r = 1+\sqrt{3}$, $|X_{i+4} - X_i| = 1$.*

*Proof.* Identifying $\mathbb{R}^2$ with $\mathbb{C}$, $X_{i+4} - X_i = r\sum_{j=i}^{i+3} e^{\mathrm{i}h_j} = r\sum_{j=0}^{3} e^{\mathrm{i}h_j}$, since the headings have period four. Now

$$\sum_{j=0}^{3} e^{\mathrm{i}h_j} \;=\; 1 - e^{-\mathrm{i}A} + e^{-\mathrm{i}(A+B)} - e^{-\mathrm{i}B} \;=\; \big(1 - e^{-\mathrm{i}A}\big)\big(1 - e^{-\mathrm{i}B}\big),$$

whose modulus is $4\sin\frac{A}{2}\sin\frac{B}{2} = 2\big(\cos\frac{B-A}{2} - \cos\frac{B+A}{2}\big) = 2(\cos\varphi_1 - \cos\varphi_d) = 2\cdot\frac{d^2-1}{2r^2} = \frac{d^2-1}{r^2}$. Multiplying by $r$ gives $|X_{i+4} - X_i| = (d^2-1)/r$. □

So for $r = \sqrt{2}$ the pairs $(X_i, X_{i+4})$, $i = 0,1,2$, are genuine $d$-edges of the two-distance graph, and for $r = 1+\sqrt{3}$ they are unit edges: the chain's long-range structure is policed by the forbidden distances even though its steps are not. To stiffen the configuration we add five *apexes* $C_1, \dots, C_5$: $C_i$ is the rotation of $X_{i-1}$ about $X_i$ through $\pm\varphi_1$ (one branch per $i$). Then $|C_iX_i| = r$ and $|C_iX_{i-1}| = 1$ automatically, and the branch is chosen so that $|C_iX_{i+1}| = d$ — exactly, for every $i$, at both radii (Section 3.3); the apexes are distinct from all chain points. The resulting configuration $Q_r = \{X_0, \dots, X_6, C_1, \dots, C_5\}$ has 12 points; its $\{1,d\}$-graph has 14 edges (5 unit and 9 at distance $d$ for $r = \sqrt{2}$; 9 and 5 respectively for $r = 1+\sqrt{3}$). The *charged slots* are the eleven designated pairs at distance $r$:

$$(X_i, X_{i+1})\ \ (0 \le i \le 5) \qquad \text{and} \qquad (X_i, C_i)\ \ (1 \le i \le 5).$$

**Proposition 3.5** (Chain caps)**.** *For $r = \sqrt{2}$ (chain A) and for $r = 1+\sqrt{3}$ (chain B): in every proper 6-coloring of $G(Q_r; \{1,d\})$, at most 5 of the 11 charged slots are monochromatic, and 5 is attained. Certificates:* `capA.cnf` *and* `capB.cnf` *(proper 6-coloring ∧ at least 6 monochromatic slots; 188 variables, 646 clauses each) are UNSAT with DRAT traces* `capA.drat`*,* `capB.drat` *verified by* `drat-trim`*; the optima were computed by MaxSAT (RC2/CaDiCaL) and confirmed by exhaustive enumeration — on the exact edge lists of Section 3.3.*

By Lemma 2.2 with the eleven slots (all at distance $r$) and $t_{\max} = 5$:

**Corollary 3.6.** *$11\,p_{\sqrt{2}} \le 5$ and $11\,p_{1+\sqrt{3}} \le 5$; that is, $p_{\sqrt{2}} \le \frac{5}{11}$ and $p_{1+\sqrt{3}} \le \frac{5}{11}$.*

This is, to our knowledge, the first *finite strict-half* bound $p_r \le \frac{1}{2} - \varepsilon$ established for any number of colors $k \ge 5$: a single 12-point configuration forcing slot mono-density $\le \frac{5}{11} = \frac{1}{2} - \frac{1}{22}$. Parts' sketch [11] obtains $p_r < \frac{1}{2}$ by counting along arbitrarily long zigzag chains; our reconstruction of that argument yields $p_r \le \frac{1}{2} - \frac{1}{188}$ (plain zigzag, chain constant $C(6) = 94$), improved to $\frac{1}{2} - \frac{1}{146}$ by the mixed-heading word above, while the finite 12-point form with apexes reaches $\frac{1}{2} - \frac{1}{22}$ and is exactly optimal for its configuration.

### 3.3 Exact-arithmetic certification of the chains

Every geometric statement about $Q_{\sqrt{2}}$ (chain $A$) and $Q_{1+\sqrt{3}}$ (chain $B$) — including the apex placements — has been verified in exact symbolic arithmetic; no floating point enters the $\chi \geq 7$ certificates. The verifier is `exactverify/verify_exact.py` (∼250 lines, on `sympy` + `python-sat`); its design:

- *Towers.* All coordinates of chain $A$ lie in $K_A = \mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{7}, S)$ with $S = \sqrt{9+4\sqrt{3}}$ (note $\sin\varphi_d = S/4$, $\sin\varphi_1 = \sqrt{7}/4$ at $r = \sqrt{2}$), a tower of nominal degree 16; those of chain $B$ lie in $K_B = \mathbb{Q}(\sqrt{3}, \sqrt{7}, T)$ with $T = \sqrt{9-4\sqrt{3}}$ (here $\sin\varphi_1 = T/4$, $\sin\varphi_d = \sqrt{7}/4$), of degree 8. Elements are polynomials in the generators, reduced to normal form modulo the defining relations $S^2 = 9+4\sqrt{3}$, $w_2^2 = 2$, $w_7^2 = 7$, $w_3^2 = 3$ (resp. $T^2 = 9-4\sqrt{3}$, …). Under a lexicographic order the relation set has pairwise-coprime pure-square leading monomials, hence is a Gröbner basis, so normal forms are unique and the quotient ring has dimension 16 (resp. 8).

- *Soundness certificate.* Nonzero normal form certifies a nonzero field element only if the quotient ring *is* the field. This is certified separately: the minimal polynomial of the primitive element $\sqrt{2}+\sqrt{3}+\sqrt{7}+\sqrt{9+4\sqrt{3}}$ (resp. $\sqrt{3}+\sqrt{7}+\sqrt{9-4\sqrt{3}}$) is computed and its degree confirmed to be 16 (resp. 8). Every subsequent equality/inequality decision is therefore exact and unconditional.

- *Checks.* For each chain: the headings are exact unit vectors; all six steps satisfy $|X_iX_{i+1}|^2 = r^2$ exactly; the five apexes satisfy $|C_iX_i|^2 = r^2$, $|C_iX_{i-1}|^2 = 1$, and $|C_iX_{i+1}|^2 = d^2 = 2+\sqrt{3}$ exactly; the three long pairs satisfy $|X_iX_{i+4}|^2 = (d^2-1)^2/r^2$ exactly ($= 2+\sqrt{3}$ for chain $A$, $= 1$ for chain $B$), confirming Lemma 3.4 independently; and *all* 66 vertex pairs are classified exactly, reproducing the 14-edge sets used by the SAT certificates and confirming that all 11 charged slots are exactly at distance $r$ and are non-edges. The exact coordinates agree with the published float coordinates (`chain12.json`, `chain12b.json`) to $10^{-9}$.

- *Caps on exact edges.* Proposition 3.5 is re-proved on the exact edge lists by exhaustive enumeration: all $\binom{11}{6} = 462$ six-subsets of the slots are unsatisfiable as simultaneous monochromatic sets under proper 6-coloring, while a five-subset is satisfiable; hence the maximum is exactly 5.

### 3.4 Proof of Theorem 1.1

*Proof of Theorem 1.1.* Suppose $c$ is a proper 6-coloring of $G(\mathbb{R}^2; \{1, d\})$ and fix an invariant mean $\Lambda$ (ZFC). By Corollary 3.3, $p_{\sqrt{2}} + p_{1+\sqrt{3}} \geq 1$; by Corollary 3.6, $p_{\sqrt{2}} + p_{1+\sqrt{3}} \leq \frac{10}{11}$. Contradiction. Hence no proper 6-coloring exists and $\chi\big(\mathbb{R}^2; \{1, d\}\big) \geq 7$. □

### 3.5 The framework caveat, honestly

Two caveats separate Theorem 1.1 from a classical finite-witness bound such as de Grey's $\chi(\mathbb{R}^2) \geq 5$. (A third — floating-point placement of the chain apexes — was eliminated by the exact-arithmetic certification of Section 3.3: the V8 and both chains are now verified exactly, end to end.)

(i) *Nonconstructive averaging.* The densities $p_r$ exist by an invariant mean on the amenable group $E(2)$, whose existence uses Hahn–Banach. This is a ZFC theorem, so Theorem 1.1 rules out *arbitrary* (not merely measurable) 6-colorings in ZFC. But in ZF+DC the mean need not exist, and Shelah–Soifer [15] showed chromatic numbers of distance graphs can genuinely depend on the ambient set theory. The choice-free content of this section is exactly its three finite certificates.

(ii) *No explicit finite witness yet.* By the de Bruijn–Erdős theorem a finite 7-chromatic $\{1, d\}$-graph exists, but the argument does not exhibit one. Section 4 makes the gap precise, reduces it to a combinatorial design problem — and Section 4.3 shows the most obvious design vocabulary is arithmetically obstructed.

# 4 Toward a fully finite witness

## 4.1 The counting blueprint

The two halves of the proof are pointwise statements about finite configurations; only the step from "every placement" to "density" uses the invariant mean. A single finite graph would suffice if the slots of many chain copies could be made to *coincide* with the diagonals of many V8 copies:

**Proposition 4.1** (Uniform-multiplicity blueprint)**.** *Suppose a finite point set $W$ contains $N$ V8 copies and a family $\mathcal{C}$ of chain copies (of types A and B) such that the multiset of charged slots of $\mathcal{C}$ covers the $4N$ diagonal pairs of the V8s, each with the same multiplicity $m \geq 1$, and contains no other pairs. Then $G(W; \{1, d\})$ is not 6-colorable.*

*Proof.* Let $M$ be the number of monochromatic diagonals in a putative proper 6-coloring. Summing Proposition 3.2 over the $N$ V8s gives $M \geq 2N$. Each chain contributes 11 slots of which at most 5 are monochromatic (Proposition 3.5); summing over $\mathcal{C}$ and using the uniform multiplicity, $mM \leq \frac{5}{11} \cdot 4Nm$, i.e. $M \leq \frac{20}{11}N < 2N$. Contradiction. □

The design problem is thus to realize such an incidence structure with actual plane geometry — every slot at distance $\sqrt{2}$ or $1 + \sqrt{3}$ being simultaneously a diagonal of some unit or outer square whose V8 completion is present. A relaxed, defect-tolerant version was worked out in the session memos:

**Theorem 4.2** (Assembly theorem; choice-free finitization — announced)**.** *Let a configuration consist of $N$ copies of each chain family with slot weights $w_e = (4, 8, 9, 9, 8, 4)$ on edge slots and $w_a = (5, 7, 7, 7, 5)$ on apex slots (total weight 73), together with reflection conflict-pairings with defect $\eta$ and V8 completions on all slots with goodness defect $\theta$. If*

$$99\,\theta + 111\,\eta < 1,$$

*the configuration is not 6-colorable. No amenability and no choice are used.*

The proof (a weighted double count refining Proposition 4.1) lives in the session memos and should be written out in full before publication. The catch is quantitative: a naive faithful closure of the required incidences is Følner-dominated and balloons to roughly $10^{40}$ vertices unless local *rational-angle cap gadgets* exist. This motivates:

**Question 4.3** (Gatekeeper question)**.** Does a finite $\{1, d\}$-graph exist containing two designated $r$-pairs in a rational-angle relative position such that no proper 6-coloring makes both pairs monochromatic? The sweet spot is "not both diagonals of one unit square monochromatic" (90°, self-supplied by V8s).

The answer controls whether the finite witness has $\sim 10^3$ or $\sim 10^{40}$ vertices. Current evidence is negative at small scale: in a 1837-point, 16,000-edge closure of the $\zeta_{24}$ lattice (a 12-fold wheel of De Neve-style $S$-graphs about a hub, whose output ring is a regular 12-gon with side exactly $\sqrt{2}$ and 60° chords exactly $1+\sqrt{3}$, hence self-seeding), colorings making both diagonals of a designated unit square monochromatic exist and are found quickly. The same test inside the full 24,881-point universe is also negative: growing neighborhoods of a designated unit square through eight rounds (to 1837 points and 16,116 edges) never destroys both-diagonals-monochromatic 6-colorings. Within the tested horizon, no rational-angle cap gadget exists.

## 4.2 Obstruction theorems

The session also established, by exhaustive SAT analysis over the 6-color $S$-graph calculus, that several cheaper routes are closed:

- *Multiplicity-1:* two copies of a configuration yield at most one useful clause; naive duplication does not amplify.
- *Trees never force:* tree-like slot incidence structures cannot force a monochromatic pair.
- *No odd spindle cycles* exist among the relevant angles: no integer relation of weight $\leq 7$ holds among $\alpha = \arccos\frac{2+\sqrt{3}}{4}$, $\beta = \arccos\frac{3}{4}$, $\gamma = \arccos\frac{2-\sqrt{3}}{4}$, $\delta' = \arccos\frac{3\sqrt{6}-\sqrt{2}}{8}$, and 30° (verified computationally).
- *Two-V8 sharing is logically transparent:* for two V8s sharing 4 points, all $121 = 11^2$ joint weight-$\geq 2$ diagonal patterns are feasible; pattern-level (NAND-calculus) reasoning over V8 arrays cannot force anything.

Consequently color-flow arguments or chain-cap gadgets (as in Section 3.2) are *mandatory* ingredients for any finite witness; there is no purely pattern-logical shortcut.

## 4.3 The design-problem verdict: an arithmetic obstruction

A second wave of computations settled whether the chain/V8 vocabulary *by itself* can realize the blueprint. The verdict is negative, for structural rather than search-depth reasons. All computations below are exact (symbolic module arithmetic, exhaustive SAT pattern enumeration); their full write-up is pending.

- *Complete pattern polytopes.* The exact sets of achievable monochromatic-slot patterns were enumerated for the V8 (all patterns of weight $\geq 2$, none below) and for both chains (all achievable subsets of the 11 slots). These sets determine *every* valid linear inequality available to any counting argument built from these gadgets.
- *Trimmed caps.* Restricting chain $A$ to the 7-slot subfamily $\{1, 2, 3, 4, 7, 8, 9\}$ yields cap 3/7; exhaustive search over all 2047 nonempty slot subsets shows 3/7 is the best (lowest) achievable cap ratio of any trimmed chain gadget.

- *Max-1-cross.* Exact coincidence scans over the full hinge family ($A$–$A$, $B$–$B$, and the $A$–$B$ 45° hinges) show that two chain placements can share at most *one* charged slot, and the resulting cross-incidence graphs are bipartite. Slot multiplicity therefore cannot be accumulated by crossing chains.

- *Placement-module freeness.* The translation modules governing periodic ladder assemblies have full $\mathbb{Z}$-rank — verified exactly over $\mathbb{Q}(\sqrt{3},\sqrt{7})$ and the towers $K_A$, $K_B$ of Section 3.3, including the joint $A$–$B$ "marriage" modules under all rotation and reflection branches. The arithmetic admits no hidden rational relations, so faithful designs blow up combinatorially (box designs of size roughly $(cr)^r$ in the rank $r$).

- *Tree defect.* Within the assembly framework of Theorem 4.2, tree-shaped (acyclic-incidence) designs provably cannot win: the counting inequality tolerates a total defect budget of $1/14$, while the best tree designs leak on the order of 90%.

- *Dual emptiness.* A Farkas-dual linear program over the exact pattern polytopes characterizes precisely which gadget inventories can certify non-6-colorability of a finite assembly. With the current inventory (V8s, chains, trimmed chains, and their weighted combinations) subject to the coincidence constraints above, the dual is empty: *no* assembly of these gadgets alone, at any scale, yields a finite contradiction.

Three escape routes survive, in order of assessed promise: (i) *generic-position pigeonhole gadgets* — V8-style pigeonholes that operate in generic relative position, bypassing the coincidence arithmetic entirely (this is also the direction of Question 4.3); (ii) the *rank-one marriage door*: rational-norm vectors in the $A$-module whose squared lengths match $B$-side slot forms, the single coincidence channel the freeness computation leaves open; and (iii) *replacement cap gadgets at new radii*, e.g. from the star-polygon family of Section 7, re-anchoring the count away from $\sqrt{2}$ and $1+\sqrt{3}$.

# 5 The record six-chromatic graphs

## 5.1 A 26-vertex 6-chromatic $\{1,d\}$-graph

**Theorem 5.1.** *The* 26 *points of Table 1 span* 85 *pairs at distance exactly* 1 *(*48 *pairs) or exactly* $d$ *(*37 *pairs), and the graph on these pairs has chromatic number* 6 *and is vertex-critical. Hence* $\chi\big(\mathbb{R}^2;\{1,d\}\big)\geq 6$ *via a* 26*-vertex witness — the smallest known* 6*-chromatic two-distance graph for any ratio.*

Each vertex is an integer 8-vector $[a_0,\dots,a_7]$ encoding the point

$$\left(\tfrac{a_0+a_1\sqrt{3}+a_2\sqrt{7}+a_3\sqrt{21}}{8},\ \tfrac{a_4+a_5\sqrt{3}+a_6\sqrt{7}+a_7\sqrt{21}}{8}\right)\in\mathbb{Q}(\sqrt{3},\sqrt{7})^2.$$

Distances are verified in exact integer arithmetic: writing a coordinate difference as $(p,q,r,s)$ in the basis $(1,\sqrt{3},\sqrt{7},\sqrt{21})/8$, its square contributes

$$\big(p^2+3q^2+7r^2+21s^2,\ 2pq+14rs,\ 2pr+6qs,\ 2ps+2qr\big)$$

in the basis $(1,\sqrt{3},\sqrt{7},\sqrt{21})/64$; summing the $x$ and $y$ contributions, a pair is a unit edge iff the total is $(64,0,0,0)$ and a $d$-edge iff it is $(128,64,0,0)$, i.e. squared distance $2+\sqrt{3}=d^2$.

| Vertices of the 26-vertex graph $([a_0,a_1,a_2,a_3 \mid a_4,a_5,a_6,a_7])$ | |
|---|---|
| $[-11,0,0,1 \mid 0,3,1,0]$ | $[-3,3,1,1 \mid 3,3,1,-1]$ |
| $[-8,-1,1,0 \mid -1,0,0,-1]$ | $[-3,4,0,1 \mid -4,3,1,0]$ |
| $[-8,0,0,0 \mid 0,0,0,0]$ | $[-3,4,0,1 \mid 4,3,1,0]$ |
| $[-8,3,-1,0 \mid -3,0,0,-1]$ | $[-3,4,2,1 \mid 2,3,1,0]$ |
| $[-8,3,1,0 \mid -5,0,0,-1]$ | $[-2,3,-1,0 \mid -3,0,-2,-1]$ |
| $[-8,3,1,0 \mid 3,0,0,-1]$ | $[0,3,-1,0 \mid -3,0,0,-1]$ |
| $[-7,0,0,1 \mid 0,-1,1,0]$ | $[0,3,1,0 \mid -5,0,0,-1]$ |
| $[-7,4,0,1 \mid -4,-1,1,0]$ | $[0,3,1,0 \mid 3,0,0,-1]$ |
| $[-6,0,0,0 \mid 0,0,2,0]$ | $[0,4,0,0 \mid -4,0,0,0]$ |
| $[-6,1,1,0 \mid -1,0,2,1]$ | $[0,7,1,0 \mid -1,0,0,-1]$ |
| $[-6,3,-1,0 \mid -3,0,2,-1]$ | $[1,4,0,1 \mid -4,-1,1,0]$ |
| $[-6,4,0,0 \mid -4,0,2,0]$ | |
| $[-5,3,1,1 \mid 3,3,-1,-1]$ | |
| $[-4,3,1,0 \mid 3,4,0,-1]$ | |
| $[-3,0,0,1 \mid 0,3,1,0]$ | |

Table 1: Integer coordinate vectors of the 26-vertex 6-chromatic $\{1,(\sqrt{6}+\sqrt{2})/2\}$-graph. Point encoded by $\big((a_0+a_1\sqrt{3}+a_2\sqrt{7}+a_3\sqrt{21})/8,\ (a_4+a_5\sqrt{3}+a_6\sqrt{7}+a_7\sqrt{21})/8\big)$.

### 5.2 A 30-vertex 6-chromatic $\{1,\sqrt{3}\}$-graph

**Theorem 5.2.** *The* 30 *points of Table 2 span* 100 *pairs at distance exactly* 1 *(*72*) or exactly* $\sqrt{3}$ *(*28*), and the graph on these pairs has chromatic number* 6 *and is vertex-critical. Hence* $\chi\big(\mathbb{R}^2;\{1,\sqrt{3}\}\big)\geq 6$ *via a* 30*-vertex witness (previously* 33 *[4]).*

Here each vertex is an integer 4-vector $[a_0,a_1,a_2,a_3]$ encoding

$$\left(\tfrac{a_0+a_1\sqrt{5}}{8},\ \tfrac{a_2\sqrt{3}+a_3\sqrt{15}}{8}\right)\in\mathbb{Q}(\sqrt{3},\sqrt{5})^2,\qquad a_0\equiv a_1,\ a_2\equiv a_3\pmod 2.$$

For a difference vector $(p,q,r,s)$, $64\cdot\mathrm{dist}^2=p^2+5q^2+3r^2+15s^2+(2pq+6rs)\sqrt{5}$, so a unit edge means exactly $(64,0)$ and a $\sqrt{3}$-edge exactly $(192,0)$.

| Vertices of the 30-vertex graph $([a_0,a_1,a_2,a_3])$ | | |
|---|---|---|
| $[-6,0,-2,0]$ | $[-1,-3,-1,1]$ | $[1,3,-1,1]$ |
| $[-5,-3,-1,1]$ | $[-1,-3,1,3]$ | $[1,3,3,1]$ |
| $[-5,3,-3,1]$ | $[-1,3,-1,3]$ | $[2,0,-2,0]$ |
| $[-4,0,-2,2]$ | $[-1,3,1,1]$ | $[2,0,0,2]$ |
| $[-4,0,0,0]$ | $[-1,3,3,3]$ | $[2,0,2,0]$ |
| $[-4,0,0,4]$ | $[0,0,-2,2]$ | $[3,-3,1,3]$ |
| $[-3,-3,-1,3]$ | $[0,0,0,0]$ | $[3,3,-1,3]$ |
| $[-3,-3,1,1]$ | $[0,0,2,2]$ | $[4,0,0,4]$ |
| $[-3,3,-1,1]$ | $[1,-3,-3,1]$ | |
| $[-2,0,-2,4]$ | $[1,-3,1,1]$ | |
| $[-2,0,0,2]$ | | |
| $[-1,-3,-3,3]$ | | |

Table 2: Integer coordinate vectors of the 30-vertex 6-chromatic $\{1,\sqrt{3}\}$-graph. Point encoded by $\big((a_0+a_1\sqrt{5})/8,\ (a_2\sqrt{3}+a_3\sqrt{15})/8\big)$.

A second, distinct 30-vertex witness (with 99 edges, differing in one vertex) was found by an independent run; both are vertex-critical.

### 5.3 Method: SAT-guided iterated local search over module closures

Both graphs were found by the same pipeline, seeded from the published coordinate tables of De Neve et al. [4] (their 33-vertex $\{1, \sqrt{3}\}$-graph and 117-vertex $\{1, d\}$-graph).

- *Universe.* A bounded-coefficient slice of the $\mathbb{Z}$-module closure of the seed coordinates: all points $p + q - r$ over the current point set staying in the same coset of the coordinate module (for $\{1, \sqrt{3}\}$: $\{((a_0 + a_1\sqrt{5})/8, (a_2\sqrt{3} + a_3\sqrt{15})/8) : a_i \in \mathbb{Z},\ a_0 \equiv a_1,\ a_2 \equiv a_3 \pmod 2\}$, $\approx 10^4$ points within radius 3.4). The closure contains coefficient patterns the seed constructions never generate; the 26-vertex graph uses several such points.

- *Oracle.* An incremental CaDiCaL instance with per-vertex selector assumptions answers "is this subset still 6-chromatic?" in microseconds.

- *Search.* Randomized greedy vertex deletion descends to critical subgraphs; perturbation kicks (adjacency-weighted candidate insertion, remove-$k$, re-descend) escape local minima. The 30-vertex minimum was reached within $\sim 10^4$ oracle calls ($\approx$ 1 minute on an Apple M4 Pro); it shares 23 vertices with the 33-vertex seed, omits 10, and adds 7 new. The $117 \to 26$ collapse additionally exploited a *pinned-anchor artifact*: the search's symmetry-breaking clique had been exempt from deletion, and an unpinned post-pass cascaded $29 \to 26$.

### 5.4 Verification recipe

Every claim in Theorems 5.1 and 5.2 is re-derivable from the integer vector tables alone:

1. *Exact edges.* Recompute all pairwise squared distances in integer arithmetic as above; no floating point enters any certificate.

2. *Non-5-colorability.* The direct 130– resp. 150-variable CNF (one variable per vertex-color pair, no symmetry-breaking assumptions) is UNSAT, proven independently by CaDiCaL 1.9.5 and Glucose 4.2, and certified by DRAT traces (`record26.drat`: an 18,518-step refutation; `record30.drat` likewise) checked with `drat-trim` (`s VERIFIED`).

3. 6-*colorability and criticality.* Explicit 6-colorings were found by SAT; all single-vertex-deleted subgraphs are 5-colorable (explicit colorings).

4. *Standalone verifiers.* `verify30.py` and `verify117best.py` ($\sim$ 60 lines each; dependency: `python-sat`) recompute the edge sets exactly from the integer vectors and re-solve everything from scratch in seconds.

*Remark* 5.3. Whether a 29-vertex (or smaller) 6-chromatic two-distance graph exists for any ratio is open. Note that the bound $v_6 \geq 42$ for unit-distance graphs [3] does not apply to two-distance graphs, where only trivial lower bounds are known for $\{1, \sqrt{3}\}$; the 26-vertex graph is vertex-critical and survived extended attack.

# 6 Unit-distance results

## 6.1 $e_5 \le 2278$

The smallest known 5-chromatic unit-distance graph has 509 vertices and 2442 edges [13, 3]. It was known not to be edge-critical: de Grey and Parts removed 36 edges non-exhaustively, giving $e_5 \le 2406$, and noted room for improvement [3].

**Theorem 6.1.** *The 509-vertex graph contains a spanning subgraph with 2278 edges (164 removed) that is not 4-colorable and is edge-critical: removing any single further edge admits a proper 4-coloring. Hence $e_5 \le 2278$.*

The trim used the same incremental selector-assumption technique as Section 5.3, over edges instead of vertices: repeatedly attempt to delete a random edge, keeping the deletion iff the remainder is still 4-uncolorable, until no edge is removable. The final 2278-edge set is certified non-4-colorable by a DRAT trace (`edge2278.cnf`/`edge2278.drat`, checked with `drat-trim`); criticality is witnessed by explicit 4-colorings of all 2278 single-edge deletions. A second, independently seeded run terminated at 2279 edges, indicating the greedy endpoint is order-dependent; the true $e_5$ may be lower still.

## 6.2 Forced structure in 5-colorings of the plane

The transfer principle of Section 2.2 applies equally with $k = 5$ and $D = \{1\}$: a non-5-colorable $\{1, d'\}$-graph whose $m$ $d'$-edges cannot all be avoided forces monochromatic pairs at distance $d'$ in *every* proper unit-distance 5-coloring of the plane (cf. Gwyn–Stavrianos [7]).

**Corollary 6.2.** *In every proper 5-coloring of $G(\mathbb{R}^2; \{1\})$, with $p_r$ the invariant densities of Definition 2.1:*

$$p_{\sqrt{3}} \ \ge \ \tfrac{1}{28}, \qquad p_d \ \ge \ \tfrac{1}{37}.$$

*Proof.* Consider the 30-point set of Theorem 5.2. Any 5-coloring of the plane proper at distance 1 restricts to a coloring of the 30 points proper on the 72 unit edges; since the full graph is not 5-colorable, at least $t = 1$ of the 28 $\sqrt{3}$-pairs is monochromatic. (MaxSAT shows $t = 1$ exactly: two cannot be forced.) By Lemma 2.2, $28\, p_{\sqrt{3}} \ge 1$. The same argument on the 26-point set (Theorem 5.1; 37 $d$-pairs, $t = 1$) gives $37\, p_d \ge 1$. □

These are the strongest known single-certificate forced-structure constants for 5-colorings; the previous implicit best, via Parts' 397-vertex $\{1, 2\}$-graph, was of order $1/550$. Conversely $p_{d'} \le \frac{1}{2}$ for all $d' \ge \frac{1}{2}$ by a color-count-free spindle argument, so any construction achieving $t/m > \frac{1}{2}$ would prove $\chi(\mathbb{R}^2) \ge 6$; the $t/m$ ladder is a quantitative measure of progress toward the fundamental lower bound, and Section 7 lowers the winning threshold from $\frac{1}{2}$ to $\frac{1}{3}$ at the right radius. (The pursuit of small $m$ is what produced the 26-vertex graph: the session's minimization runs plateaued at $m = 37$ $d$-edges.)

## 6.3 An automated density ledger and $p_{\sqrt{3}} \ge 1/12$

Single certificates are only the first rung. The session built an automated *density ledger*: approximately 143,000 MaxSAT-exact linear inequalities over monochromatic-pair densities, mined from small unit-rigid configurations (each inequality of the form $\sum_i \alpha_i p_{r_i} \ge t$ or $\le t$, certified as in Lemma 2.2), coupled with triangle relations among densities and solved as a linear program.

**Theorem 6.3** (Density ledger — announced)**.** *For every isometry-invariant proper* 5*-coloring of the plane,* $p_{\sqrt{3}} \geq \frac{1}{12}$*. The LP also reproduces the classical hexagonal-packing inequality as a special case.*

The LP is feasible — the ledger is a quantitative map of how much freedom 5-colorings retain — and the framework turns "prove $\chi(\mathbb{R}^2) \geq 6$" into the concrete task "find configurations whose inequalities make this LP infeasible."

### 6.4 The 5-coloring wall, measured

Is any *single* distance forced monochromatic inside an actual 5-chromatic graph? A sweep over Heule's 803-vertex 5-chromatic graph [8] tested, for each of the 428 distance classes with at least 40 point pairs, whether some proper 5-coloring avoids monochromatic pairs in that class entirely. The answer is yes for every class — no forced single-distance family exists in this graph — but the classes are far from equal: every class resolved in milliseconds to seconds, except $d = 2$, which required 3.4 hours of solver time before a witness coloring was found. Distance 2 — precisely the distance the two-distance theory singles out (Exoo–Ismailescu's $\{1, 2\}$ six-chromatic construction, and $d^2 - 1$-type coincidences) — is where the 5-coloring solution space is thinnest.

## 7 Strict caps for five colors and a win condition

*(Superseded: this section was written before [18] proved* $\chi(\mathbb{R}^2) \geq 6$*. We keep it for the cap theorems, which remain valid.)*

Corollary 3.6 caps a density strictly below $\frac{1}{2}$ for 6-colorings. This section develops the analogous theory on the unit-distance side: caps valid for colorings with *any* number of colors, an impossibility theorem explaining where such caps cannot exist, and the resulting explicit win condition for $\chi(\mathbb{R}^2) \geq 6$. Throughout, "proper" means proper at distance 1, and $p_r$ is the invariant density of Definition 2.1 for such a coloring.

### 7.1 Color-count-free finite caps

**Proposition 7.1** (Triangle and circumcenter)**.** *Let* $Q$ *consist of a unit equilateral triangle* $v_1v_2v_3$ *and its circumcenter* $o$*, with* $|o - v_i| = 1/\sqrt{3}$*. In every proper coloring (any number of colors), at most one of the three pairs* $(o, v_i)$ *is monochromatic. Hence* $p_{1/\sqrt{3}} \leq \frac{1}{3}$ *for every proper coloring of the plane.*

*Proof.* The $v_i$ are pairwise at distance 1, so they receive three distinct colors; $o$ can agree with at most one of them. Transfer (Lemma 2.2) gives $3\,p_{1/\sqrt{3}} \leq 1$. □

**Proposition 7.2** (Unit pentagon)**.** *Let* $\tau = (1+\sqrt{5})/2$*. The regular pentagon with unit sides has diagonals of length* $\tau$*; in every proper coloring, at most two of the five diagonals are monochromatic, so* $p_\tau \leq \frac{2}{5}$*. Dually, the regular pentagon with unit diagonals has sides of length* $1/\tau$ *and caps* $p_{1/\tau} \leq \frac{2}{5}$*.*

*Proof.* Label the vertices $v_0, \ldots, v_4$ cyclically. If two diagonals sharing a vertex, say $(v_0, v_2)$ and $(v_2, v_4)$, were both monochromatic, then $c(v_0) = c(v_4)$, contradicting the unit side $(v_4, v_0)$. So monochromatic diagonals form an independent set in the diagonal 5-cycle $v_0v_2v_4v_1v_3$, of size at most 2. The dual statement is identical with sides and diagonals exchanged. □

**Proposition 7.3** (Odd stars and flowers)**.** *For odd $n$ and $1 \le a < n/2$ with $\gcd(a,n) = 1$: place a regular $n$-gon so that its step-$2a$ chords have length 1; its step-$a$ chords then have length $r = 1/(2\cos(a\pi/n))$, and in every proper coloring at most $(n-1)/2$ of the $n$ step-$a$ chords are monochromatic: $p_r \le \frac{n-1}{2n} < \frac{1}{2}$. Similarly the "flower" (regular $m$-gon with unit sides plus its center, $m$ odd) caps $p_r \le \frac{m-1}{2m}$ at the circumradius $r = 1/(2\sin(\pi/m))$; $m = 3$ recovers Proposition 7.1. The capped radii $1/(2\cos(a\pi/n))$ are dense in $(\frac{1}{2}, \infty)$.*

*Proof sketch.* Two step-$a$ chords sharing a vertex, $(v_i, v_{i+a})$ and $(v_{i+a}, v_{i+2a})$, cannot both be monochromatic: that would force $c(v_i) = c(v_{i+2a})$ across the unit step-$2a$ chord. So monochromatic step-$a$ chords are independent in the step-$a$ cycle, which is a single *odd* cycle since $\gcd(a,n) = 1$ and $n$ is odd; an odd $n$-cycle has independence number $(n-1)/2$. The flower argument is the same with spokes in place of chords. □

These are, to our knowledge, the first finite sub-half density caps relevant to $k \ge 5$ — and they are stronger than needed: they hold for arbitrary proper colorings, with no bound on the number of colors.

## 7.2 An impossibility theorem at the ledger anchors

Why do the caps above live at $1/\sqrt{3}$, $\tau$, $1/(2\cos(a\pi/n))$ and not at the density-ledger anchors $\sqrt{3}$ and 2 where they are most wanted? All of them run on the same fuel: two $r$-slots sharing an endpoint conflict when their far endpoints are at distance 1, and the angle subtended at the shared endpoint is then $\varphi_1(r) = 2\arcsin\frac{1}{2r}$, with $\cos\varphi_1 = 1 - \frac{1}{2r^2}$. A cap needs *odd* conflict cycles, i.e. closed walks of such conflicts — and those exist only when $\varphi_1$ is commensurable with $\pi$ (at $r = 1/\sqrt{3}$: $\cos\varphi_1 = -\frac{1}{2}$, $\varphi_1 = 2\pi/3$; at $r = \tau$: $\cos\varphi_1 = \tau/2$, $\varphi_1 = \pi/5$).

**Theorem 7.4** (Pairwise impossibility at $\sqrt{3}$ and 2 — announced)**.** *At $r = \sqrt{3}$ and $r = 2$ the conflict angle has rational cosine ($\cos\varphi_1 = \frac{5}{6}$ and $\frac{7}{8}$ respectively), so by Niven's theorem [10] $\varphi_1$ is an irrational multiple of $\pi$. Consequently all conflict cycles are even, every pairwise-conflict graph on $r$-slots is bipartite, and the optimal pairwise cap is exactly $\frac{1}{2}$: no finite configuration whose only mechanism is pairwise slot conflict can prove $p_{\sqrt{3}} < \frac{1}{2}$ or $p_2 < \frac{1}{2}$.*

The bipartite bound is tight in practice: SAT verification through 45-point scaffolds, and exhaustive chain scans at $r = \sqrt{3}$ and $r = 2$ (all apex/pinch turn words, with and without waist points), all plateau at density exactly $\frac{1}{2}$ — while the same scans at $r = 1/\sqrt{3}$ immediately break $\frac{1}{2}$ (best chain found: 3/8). Beating $\frac{1}{2}$ at $\sqrt{3}$ or 2 therefore requires non-pairwise, de Grey-scale machinery.

## 7.3 The win condition

The caps convert record-graph engineering into a quantitative attack on $\chi(\mathbb{R}^2) \ge 6$.

**Theorem 7.5** (Win condition)**.** *Let $G$ be a finite $\{1, \sqrt{3}\}$-graph with $m$ unit edges such that every 5-coloring of its vertices that is proper on the $\sqrt{3}$-edges makes at least $t$ unit edges monochromatic (any 6-chromatic $\{1, \sqrt{3}\}$-graph has $t \ge 1$). If $t/m > \frac{1}{3}$, then $\chi(\mathbb{R}^2) \ge 6$.*

*Proof.* Suppose $c$ is a proper 5-coloring of $G(\mathbb{R}^2; \{1\})$. Scale $G$ by $1/\sqrt{3}$: its $\sqrt{3}$-edges become unit pairs and its unit edges become pairs at distance $1/\sqrt{3}$. For every isometric placement of the scaled vertex set, $c$ restricted to it is a 5-coloring proper on the former $\sqrt{3}$-edges, so by hypothesis at least $t$ of the $m$ former unit edges are monochromatic. Lemma 2.2 gives $p_{1/\sqrt{3}} \ge t/m > \frac{1}{3}$, contradicting Proposition 7.1. Hence no proper 5-coloring exists. □

The current certified value of the ratio is $t/m = 1/72$: the 30-vertex graph of Theorem 5.2 has 72 unit edges and forces $t \geq 1$ (immediate from non-5-colorability; MaxSAT shows $t = 1$ exactly). A direct optimizer over the $\mathbb{Q}(\sqrt{3}, \sqrt{5})$ module universe — iterated local search minimizing the unit-edge count among 6-chromatic $\{1, \sqrt{3}\}$-graphs — has already reached $m = 70$ on 30 vertices (ratio 1/70, pending exact re-verification). The golden-ratio variant uses Proposition 7.2: a 6-chromatic $\{1, \tau\}$-graph forcing unit-edge monochromatic density above $\frac{2}{5}$ under $\tau$-proper 5-colorings also proves $\chi(\mathbb{R}^2) \geq 6$; Parts' 31-vertex graph [12] measures 1/57 against that threshold. The gap between 1/70 and 1/3 is large, but Theorem 7.5 makes it a single, well-posed finite optimization — raise $t$ (multi-fold forcing) and lower $m$ — with no invariant-mean caveats attached to the prize: by Theorem 7.4 the route through the caps at $1/\sqrt{3}$ and $1/\tau$ is the only pairwise one available.

## 8 An extended six-coloring continuum

Following Mundinger, Pokutta, Spiegel and Zimmer [21], let $X_6$ be the set of $d > 0$ for which the plane has a coloring with six colors in which five color classes contain no two points at distance 1 and the sixth contains no two points at distance $d$. They showed $X_6 \supseteq [0.354, 0.657]$ (to three decimals). Reaching $d = 1$ would prove $\chi(\mathbb{R}^2) \leq 6$.

**Theorem 8.1.** $X_6 \supseteq [0.354,\ 0.667504532081]$.

The new part is $[657/1000,\ 667504532081/10^{12}]$. It is covered by 59 periodic tilings with rational vertex coordinates, each with eight polygonal tiles per period, from a 33-parameter family whose mirror symmetry is broken. For each tiling an exact rational-arithmetic verifier checks that all tiles are simple polygons, that periodic interiors do not overlap, that the tile areas sum exactly to the area of the lattice cell, and that every same-color diameter and separation inequality holds strictly. The last check yields an open interval of distances $d$ that the sixth color avoids. A second script checks that these 59 intervals overlap and cover the stated range. The numerical optimum of the family is $d = 0.66750453208133\ldots$. At this point 23 constraints are active with positive multipliers, and the reduced Hessian is negative definite, so it is a strict second-order local maximum of the tiling topology. Period doubling, $2 \times 2$, $3 \times 1$, $1 \times 3$ and $3 \times 2$ supercells, and all period-two relabelings of the five unit colors return to the same value. The certificates and verifiers are in the reproducibility bundle.

## 9 Discussion and open problems

**Sparsity walls.** On the negative side, the session derived quantitative lower bounds ("sparsity walls") showing that forcing results require substantial graphs. Over the Gwyn–Stavrianos framework [7], the master inequality $\sum_P q_c(d_{ab}) + t\, e\, p_5(c) \geq t$ yields: any 5-color forcing unit-distance graph (a UDG forcing a monochromatic pair at some distance $d' \geq \frac{1}{2}$ in all proper 5-colorings) has at least 21 vertices and 90 edges; for $d' = 1$, at least 32 vertices; at least 25 vertices for families in Parts' window, and $\approx 25(r+1)$ under $r$-robustness; $t$-fold forcing needs $n \geq 17.9\, t^{3/4}$ vertices. Moreover every unit-distance graph on at most 15 vertices is 5-colorable (degeneracy plus the exact maximum edge counts $u(n)$), and any 6-chromatic unit-distance certificate must have at least $\sim$6993 vertices (via Parts' bound $\delta_6 < 1.43 \times 10^{-4}$).

**Open problems.**

**Problem 9.1.** Is there a 25-vertex 6-chromatic two-distance graph? The 26-vertex graph is vertex-critical and has survived extended attack; no nontrivial lower bound on the minimum order is known for two-distance graphs.

**Problem 9.2** (Finite 7-chromatic witness)**.** Realize the incidence structure of Proposition 4.1/Theorem 4.2 by an actual finite $\{1, d\}$-point set, or answer Question 4.3. A positive answer to the gatekeeper question would put a $\sim 10^3$-vertex witness in reach; a 6-color forced-same pair (two points forced to share a color) would yield a 7-chromatic graph immediately by a spindle, at roughly $2|W|$ vertices.

**Problem 9.3** (Climbing to the win condition)**.** Push the ratio of Theorem 7.5 from 1/70 toward $\frac{1}{3}$: find 6-chromatic $\{1, \sqrt{3}\}$-graphs with few unit edges and, more importantly, with *multi-fold* forcing ($t \geq 2$) under $\sqrt{3}$-proper 5-colorings — no construction with $t \geq 2$ is currently known. By Theorem 7.4, pairwise mechanisms can never beat $\frac{1}{2}$ at the anchors $\sqrt{3}$ and 2 themselves, so progress must come either through the capped radii ($1/\sqrt{3}$, $1/\tau$, and the star-polygon family), through non-pairwise machinery at $\sqrt{3}$ or 2, or through new inequalities feeding the density-ledger LP of Section 6.3.

**Problem 9.4.** Determine $e_5$. The greedy endpoint 2278 is order-dependent (2279 on a second run); no matching lower bound technique for edge counts of 5-chromatic UDGs is known beyond the vertex-based walls above.

# A Artifacts and reproducibility

All certificates live in the bundle `hadwiger-nelson/` (session archive). Trusting any UNSAT claim requires trusting only the ∼1300-line DRAT checker `drat-trim` [17] (included as a single C file) plus integer arithmetic; trusting the SAT (colorability) claims requires only checking explicit colorings; the exact-arithmetic chain certificates additionally rest on `sympy`'s multivariate polynomial reduction and minimal-polynomial routines. The $k$-independent cap searches and the design-verdict computations (Sections 4.3 and 7) live in the session archive directories `k5caps/` and `design7/`.

| File(s) | Size | Semantics |
|---|---|---|
| `v8cert.cnf/.drat` | 2.7 KB / 262 KB | proper 6-coloring of the V8 with $\leq 1$ monoch matic diagonal: UNSAT (Proposition 3.2); vars, 267 clauses. |
| `capA.cnf/.drat` | 7.2 KB / 21 KB | proper 6-coloring of chain $A$ ($r = \sqrt{2}$) with of 11 slots monochromatic: UNSAT (Prop tion 3.5); 188 vars, 646 clauses. |
| `capB.cnf/.drat` | 7.2 KB / 21 KB | same for chain $B$ ($r = 1 + \sqrt{3}$). |
| `chain12.json, chain12b.json, v8.json` | <1 KB each | coordinates and designated pairs of the th $\chi \geq 7$ configurations. |
| `record26.cnf/.drat` | 4.9 KB / 884 KB | 5-coloring of the 26-vertex graph: UNS (18,518-step refutation). |
| `record30.cnf/.drat` | 5.8 KB / 1.3 MB | 5-coloring of the 30-vertex graph: UNSAT. |
| `edge2278.cnf/.drat` | 126 KB / 74 MB | 4-coloring of the 2278-edge subgraph: UNSA |
| `edge2278_graph.json` | 26 KB | the 2278-edge subgraph of the 509-vertex gra |
| `deneve117_min.json` | <1 KB | the 26 integer vectors (Table 1). |
| `verify30.py, verify117best.py, deneve_verify.py` | ∼60 lines each | standalone verifiers: recompute edges exa from integer vectors, re-solve 5-/6-colorabi and vertex-criticality from scratch (dependen `python-sat`). |
| `verify_exact.py` | ∼250 lines | exact symbolic certification of chains $A$ $B$ (Section 3.3): tower-degree soundness tificates via primitive-element minimal poly mials, all 66 pairwise distances per chain solved in $K_A$ (degree 16) resp. $K_B$ (deg 8), edge/slot sets confirmed against the f files, caps re-proved by exhaustive subset e meration on the exact edge lists (dependenc `sympy`, `python-sat`). |
| `drat-trim(.c)` | 60 KB | the proof checker; all `.drat` traces above ch as `s VERIFIED`. |

Reproduction: `pip install python-sat`, then `python3 verify30.py` (seconds) and `python3 verify117best.py deneve117_min.json` (seconds); for any UNSAT certificate, `./drat-trim X.cnf X.drat` and confirm `s VERIFIED`. All searches ran on a single Apple M4 Pro laptop; total compute for the record graphs was minutes, for the edge trim hours, and for the density ledger overnight.

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We thank Jaan Parts, whose 2019 sketch [11] contains the skeleton of Theorem 1.1, and De Neve, Vanden Kerchove, Colle, Tavernier, and Pickavet, whose published coordinate tables [4] seeded the searches of Section 5. Autonomous literature-retrieval agents surfaced the open-access coordinate tables at the start of the session.

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